

Nassau Candy Distributor

Factory-to-Customer Shipping Route Efficiency Analysis — Executive Summary

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Purpose: Evaluate factory-to-customer shipping route efficiency to identify the fastest and slowest performing routes, geographic bottlenecks, ship mode performance, and factory-level trends, supporting data-driven improvements to distribution efficiency.

Objective

Nassau Candy ships from five factories to customers across 49 U.S. states and Washington, D.C. This report looks at how well that shipping network actually performs — which routes run fastest, where the bottlenecks sit, and whether shipping mode or factory choice makes a measurable difference. The analysis is based on 8,389 orders (9,994 order line items) after cleaning, and the aim throughout was to surface findings a distribution team could actually act on, not just describe the data.

Data Quality Finding & Methodology

Before any of the route analysis could be trusted, the Order Date and Ship Date fields needed a closer look — and they didn't hold up. Shipping Lead Time, calculated as Ship Date minus Order Date, came out to between 904 and 1,642 days across the dataset. That's two and a half to over four years to ship candy, which isn't a real-world number under any circumstance. A few things pointed to the same conclusion:

- Order Date only spans 2024–2025, while Ship Date spans 2026–2030 — the two fields don't line up the way they would if one were genuinely derived from the other.
- Plotting Lead Time produces three sharp, isolated spikes with nothing in between — not the smooth, continuous spread a real shipping metric would show.
- Even after normalizing both dates to a common year (removing the year mismatch entirely and comparing only month and day), the resulting gap was still centered near zero, with roughly half the values negative. That ruled out a simple wrong-year explanation — the day-to-day relationship between the two dates isn't real either.
- Ship Mode adds one more data point: Same Day and First Class shipments both average a longer Lead Time than Standard Class, which is backwards from how these categories should behave.

What this means for the analysis: Lead Time isn't reliable as a literal day count, so it isn't reported that way anywhere in this summary. It's still calculated and used throughout — ranking routes, scoring efficiency, flagging delays — but always in a **relative** sense: this route compared to that one, not "customers wait X days." Everything below follows that rule. It's also worth noting this issue is specific to the two date fields; nothing here suggests Sales, Cost, geography, or product data have the same problem.

This was flagged for mentor guidance before settling on an approach. The response was that the dataset should be used as given and the problem statement followed closely — read as encouragement to work around the limitation with sound judgment rather than chase a fix that likely wasn't there, which is the approach reflected in this report.

Data Cleaning & Preparation

Two smaller decisions were made during cleaning and are worth noting for anyone reviewing the numbers:

- **U.S.-only scope:** 200 records (about 2% of the raw data) belonged to Canadian provinces. Since the project is scoped to a national U.S. distributor and the dashboard maps U.S. states, these were dropped. The working dataset is 9,994 U.S. order line items.
- **A mismatch in the Factory data:** one product, Fizzy Lifting Drinks, was labeled "Sugar" in the raw Division field, but its Product ID prefix and the official Product–Factory mapping both said "Other." Since two of the three sources agreed, the Factory assignment was built from Product Name instead of the unreliable Division field, correctly routing this product to Sugar Shack.

Headline Metrics

Metric	Value
Total Orders (unique)	8,389
Total Order Line Items	9,994
Total Sales	\$138,830.34
Total Gross Profit	\$91,508.23
Overall Delay Frequency*	33.26%
Average Route Efficiency Score	54.7 / 100

*Delay Frequency = share of orders with Lead Time in the slowest of three identified clusters ($\geq 1,634$), used as a relative delay indicator, not a literal day count.

Factory Performance

Two factories do most of the work here. Lot's O' Nuts and Wicked Choccy's handle 5,588 and 4,060 orders respectively — together, that's about 96.5% of everything shipped. That's not a red flag; it's simply because the highest-volume chocolate products happen to be mapped to these two factories. Secret Factory, The Other Factory, and Sugar Shack cover the rest (213, 100, and 33 orders).

Route Performance

177 state-level routes had enough volume (10+ orders) to compare fairly. Each was scored 0–100 on relative efficiency — 100 for the fastest route in the dataset, 0 for the slowest.

	Route	Volume	Efficiency Score
Most efficient	Wicked Choccy's → Nevada	10	100.0
Least efficient	Wicked Choccy's → New Mexico	14	0.0

High-Volume, Poor-Performance Routes

Rankings alone can be misleading — a slow route with 10 orders matters a lot less than a slow route with 500. So the more useful question is which routes are both busy and underperforming. Six qualify, and Washington shows up twice: once from Lot's O' Nuts (230 orders, score 47.3) and again from Wicked Choccy's (187 orders, score 36.9 — the worst of the six). That two different factories both struggle to serve the same state is worth attention on its own. The other four flagged routes are Lot's O' Nuts to New York (544 orders), Lot's O' Nuts to Ohio (218), Wicked Choccy's to Pennsylvania (188), and Wicked Choccy's to Illinois (183).

Geographic Bottlenecks

Looking at states rather than routes tells a similar story. Illinois, Washington, and Ohio all combine above-average order volume with above-average delay rates — Washington the worst of the three at 42.82% (compared with 35.57% for Illinois and 35.46% for Ohio, on 433 and 392 orders). Interestingly, California doesn't make this list despite having by far the most orders of any state (1,686) — its 32.92% delay rate is actually just under the cross-state average, which is being pulled up by a handful of smaller states with more erratic performance. Volume and delay clearly don't move together here.

Ship Mode Performance

Standard Class is the default for most customers — 5,026 of 8,389 orders, well ahead of Second Class (1,621), First Class (1,302), and Same Day (440). Cost doesn't vary much by mode (roughly \$4.42 to \$4.84 per shipment), so there's no real cost-speed trade-off to point to here. What is worth flagging: both First Class and Same Day average a longer Lead Time than plain Standard Class. That's the same pattern already noted in the Data Quality section, and it's further evidence that the underlying date fields aren't behaving the way real shipping data would.

Recommendations

- **Start with Washington.** It's the only state where both major factories underperform at the same time, and it has the highest state-level delay rate in the dataset (42.82%).
- **Take a closer look at Illinois and Ohio** — both show the same above-average volume, above-average delay pattern, just less extreme than Washington.
- **Don't use Ship Mode cost as a lever for speed.** The data doesn't support a cost-time trade-off, so mode selection should be driven by other operational needs.
- **Fix the Order Date/Ship Date generation issue at the source.** Until that happens, Lead Time will keep needing this relative-only treatment instead of being usable as a straightforward, trustworthy metric.